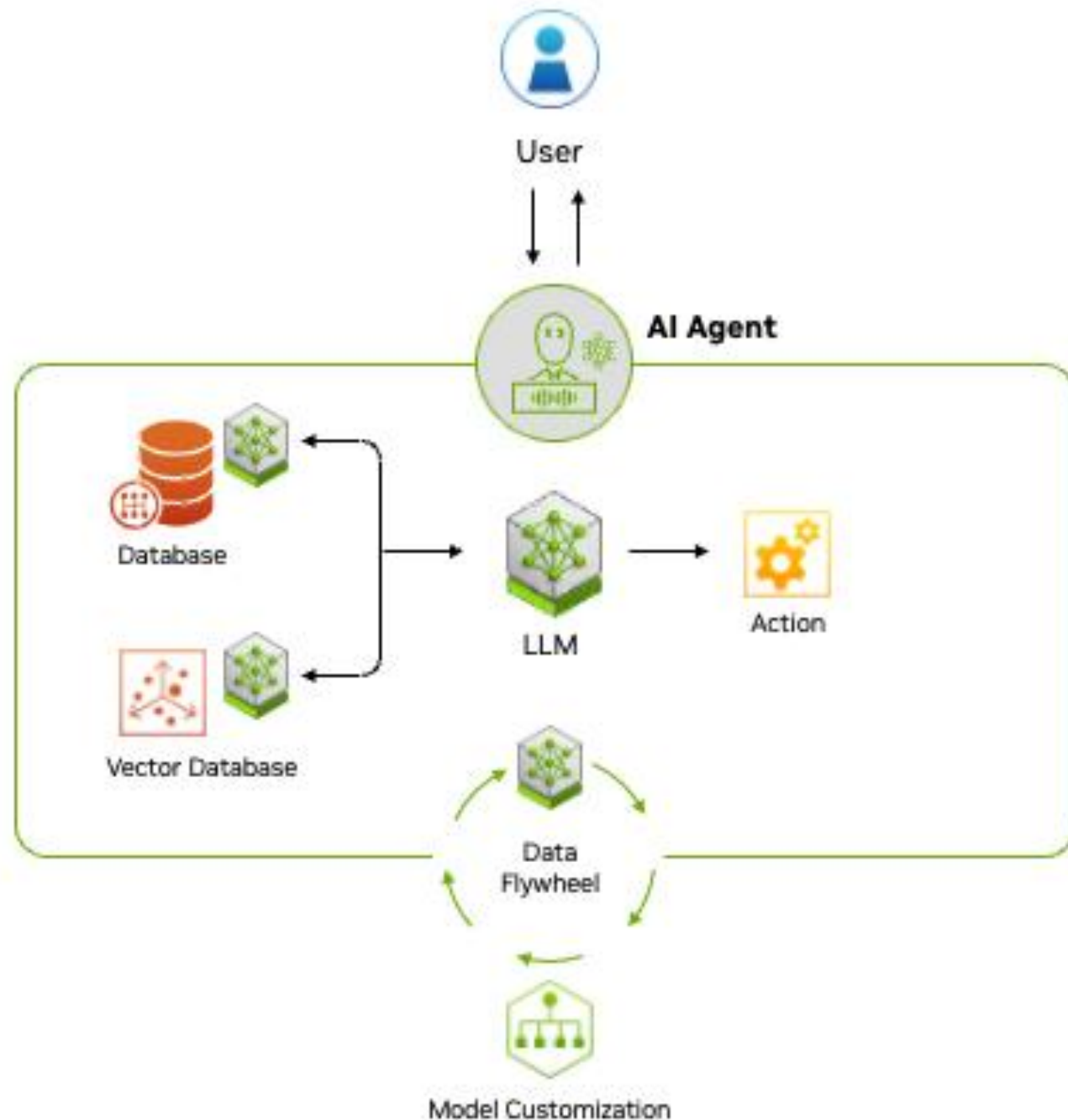
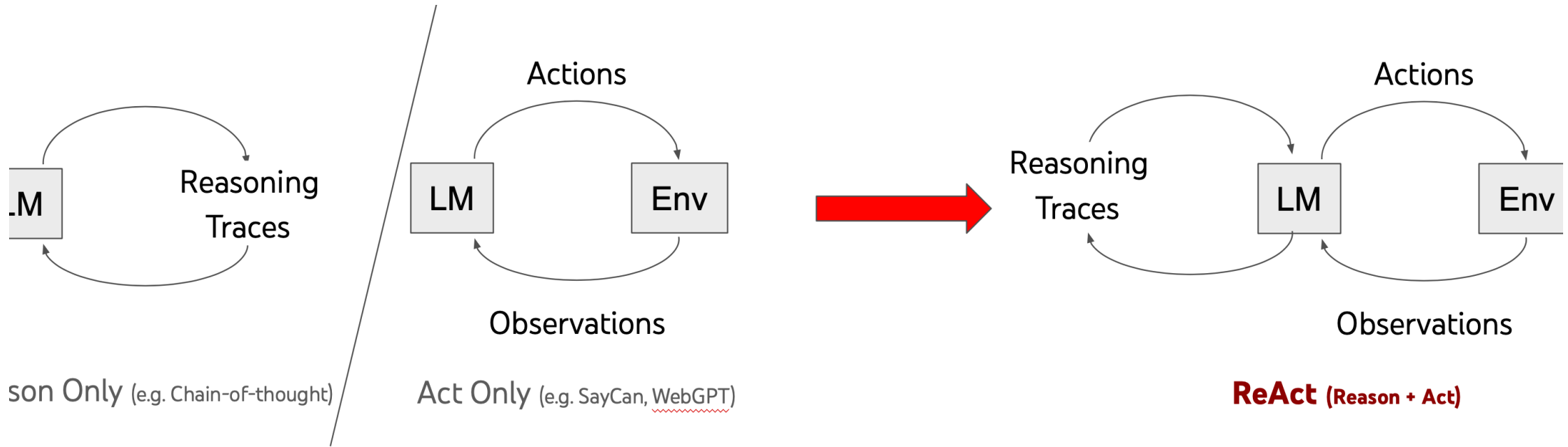


MODULE 2 - AGENTIC TOOL USE & ACTION LOGIC

- **Goal:** Moving from passive text generation to active environmental interaction.
- **Function Calling:** Developing a library of Python tools (math, search, database) that agents can autonomously "choose" to execute.
- **Tool Discovery:** Enabling agents to understand which tool fits a specific sub-task based on its description.

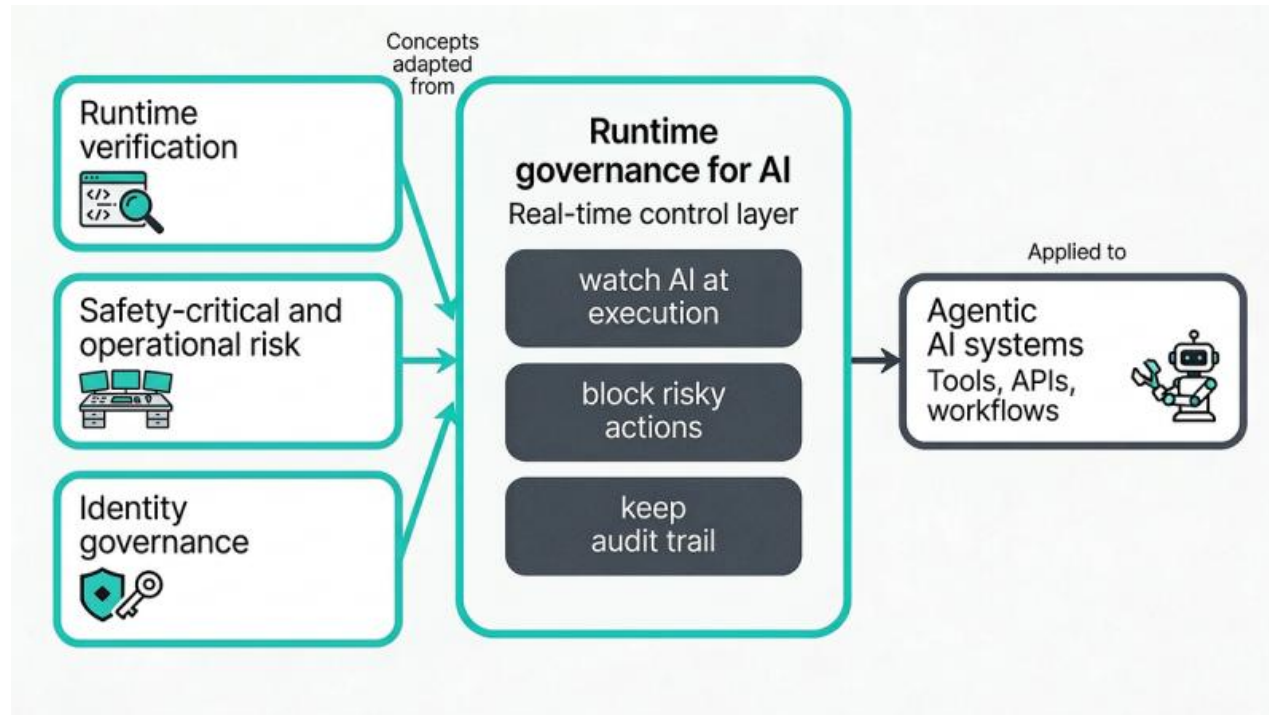




REASONING PATTERNS (REACT)

- **The Concept:** Implementing the **Reason + Act** loop.
- **Logic:** Forcing the LLM to "think" and document its reasoning before triggering a tool.
- **Example:** "I need to check the local weather before I recommend an outfit"

GOVERNANCE & SAFETY



- **Human-in-the-Loop (HITL):** Designing "Approval Nodes" where the system pauses for human confirmation before executing high-risk actions.
 - **Safe Sandbox Execution:** Running agent-generated code or terminal commands within isolated environments to prevent system damage.
 - **Task 2 Deliverable:** A script where the AI calls a custom Python function only when necessary.
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